Svistak Vladimir

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Education

Chemical Engineering Center, ITMO University 2023-2027

Experience

Practice in the Laboratory of Physical Chemistry: November 2023 – December 2024
ITMO faculty of Infochemistry

- Growing microgreens using hydroponics
- Preparation of the nutrient medium
- Dependency injection with Hilt
- Synthesis of melamine cyanurate particles
- Work in sterile conditions
- Working with a fluorescence microscope

Laboratory of Molecular Biology of Fungi: November 2024 – January 2025
Komarov Botanical Institute

- Isolation of fungal DNA by the magnet method
- Preparation of mushroom samples for DNA isolation
- Compilation of Phylogenetic trees using the maximum likelihood method
- Alignment by various BLAST algorithms
- Working with FastQC

ML prediction of aptamer-small molecule interactions: August 2025 – Present
ITMO faculty of Chemistry and AI

- Creating artificial aptamers for training (random assembly, cVAE, GRU)
- Using contrastive learning to predict the interaction between aptamers and molecules
- Testing interaction using docking, boltz, and other tools

Analysis of amino acid substitutions on the stability of amyloid proteins: December 2025 – Present
Laboratory of Amyloids of St. Petersburg State University

- Writing special. software to create gatekeeper mutations
- Docking proteins to test for adhesion
- Using ML/AI architectures to create a predictive model of amyloidogenic regions

CV in the field of detecting construction workers' helmets and safety vests: May 2026 – Present
ML Laboratory at the Financial Institute

- Using yolo models for object detection tasks
- Development of a photo submission pipeline and subsequent processing / uploading to the cloud

Technical Skills

Programming languages: Python, SQL
Frameworks and Librauries: Pandas, NumPy, Scikit-learn, Seaborn, BioPython, Matplotlib, Cv2, BeautifulSoup, requests, selenium, pyTorch, pdbfixer, MDAnalysis, freesasa, py3Dmol, openmm
Dev tools: Linux, Git, Docker, Galaxy, BEDtools, Seqtools, SAMtools, FastQC, FoldX, Botlz

Courses

Summer lecture on Bioinformatics from the Institute of Bioinformatics July 2024

* NGS sequencing and data analysis
* Fundamentals of molecular modeling
* Bioinformatics in oncology

School of Bioinformatics of the Center for Chemical Engineering May 2024

* Molecular docking
* Alignment and sequence analysis
* Parsing and reading information from various biological databases

Progects

CodonAdaptation

* Codon optimization
* Machine learning
* Molecular biology

Biokhakaton 2025

* Amyloids
* Amino acid substitutions
* Structural Biology